In 1905, Albert Einstein published an extremely important paper in which he departed from the Planckian notion that it was the absorption and emission of radiation that occurred in quanta. Einstein instead proposed that *radiation itself consisted of quanta of energy*, with the rule that the energy of a quantum of light of frequency ν is $h\nu$. Einstein then used the concept of the quantum nature of light to explain some peculiar properties of metals when these are irradiated with visible and ultraviolet light.

1-2 THE PHOTOELECTRIC EFFECT

In 1887, the photoelectric effect was discovered by Heinrich Hertz, who, while engaged in his famous experiments on electromagnetic waves, found that the length of the spark induced in the secondary circuit was reduced when the terminals of the spark gap were shielded from the ultraviolet light coming from the spark in the primary circuit. His observations attracted much interest, and the following facts were established by further experiments:

1. When polished metal plates are irradiated, they may emit electrons;[4] they do not emit positive ions.
2. Whether the plates emit electrons depends on the wavelength of the light. In general there will be a threshold that varies from metal to metal: Only light with a frequency greater than a given threshold frequency will produce a photoelectric current.
3. The magnitude of the current, when it exists, is proportional to the intensity of the light source.
4. The energy of the photoelectrons is independent of the intensity of the light source but varies linearly with the frequency of the incident light.

Although the existence of the photoelectric effect can be understood within the framework of classical electromagnetic theory, since it was known that there were electrons in metals, and one could imagine them to be accelerated by absorption of radiation, the frequency-dependence of the effect is not comprehensible within that framework. The energy carried by an electromagnetic wave is proportional to the intensity of the source, and frequency has nothing to do with it. Furthermore, a classical explanation of the effect, which would have to involve the concentration of the energy deposited on single photoelectrons, would carry with it an implied time delay between the arrival of the radiation and the departure of the electron, the delay being longer when the intensity is decreased. In fact, no such time delays were ever observed, at least none longer than 10^{-9} sec, even with incident radiation of very low intensity.

Einstein considered the implications of having the radiation consist of a collection of quanta of energy $h\nu$, where ν is the frequency of the light wave. The absorption of a single quantum of radiation by an electron—a process that may take less time than the upper limit previously quoted—increases the electron energy by an amount $h\nu$. Some of the energy must be expended to separate the electron from the metal. This amount, W (called the *work function*), might be expected to vary from metal to metal, but should not depend on the electron energy. The rest is available for the electron kinetic energy, so that on the

[4]This was established by an *e/m* measurement.

basis of this picture one expects that the following relation between electron velocity v and light frequency ν.

$$\tfrac{1}{2}mv^2 = h\nu - W \tag{1-9}$$

should hold. The threshold effect and the linear relation between electron kinetic energy and the frequency are contained in this formula. The proportionality of the current and the source intensity can also be understood in terms of these light quanta, or *photons*, as they came to be called: a more intense light source emits more photons, and these in turn can liberate more electrons.

The confirmation of Einstein's hypothesis was slow in coming. The definitive experiments carried out by the American physicist Robert A. Millikan (who did not believe in quanta!) were not completed until 1915. As Fig. 1-3 shows, they were in complete accord with (1-9). Incidentally, a byproduct of these and subsequent experiments was information about metals. It was found that W was of the order of several electron volts (1 eV is the energy required to move a charge e through one volt potential difference, and since the charge of the electron is $|e| = 1.602 \times 10^{-19}$ C, we have $1\text{ eV} = 1.602 \times 10^{-19}$ J).

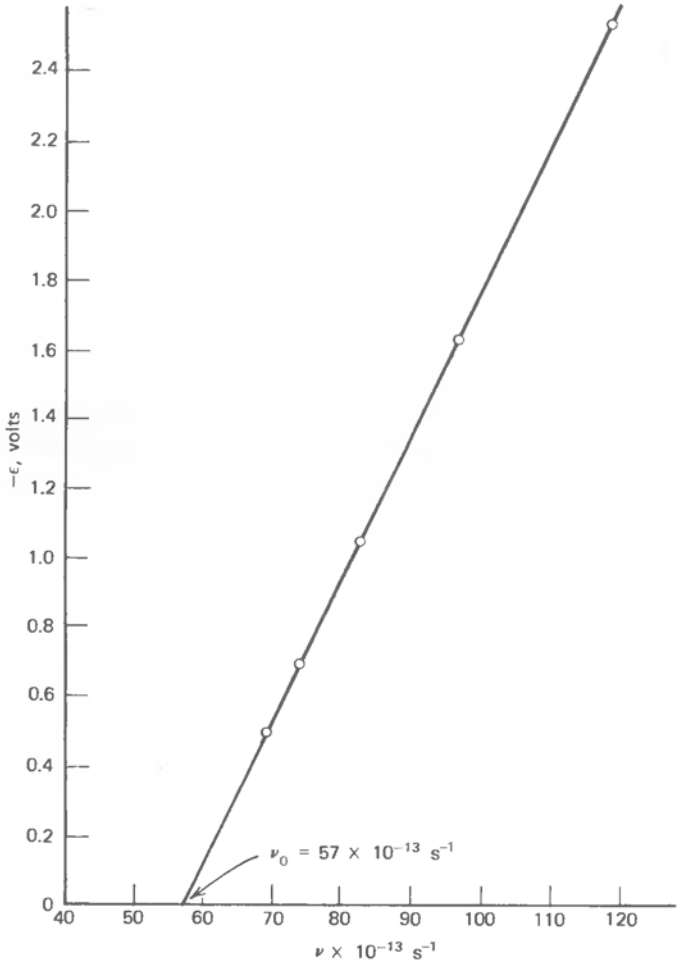


Figure 1-3 Photoelectric effect data showing a plot of retarding potential necessary to stop electron flow from a metal (lithium), or equivalently, electron kinetic energy, as a function of frequency of the incident light. The slope of the line is h/e.

EXAMPLE 1-4

Radiation of wavelength $\lambda = 290$ nm falls on a metal surface for which the work function is $W = 4.05$ eV. What potential is needed to stop the most energetic photoelectrons?

SOLUTION The energy of the photons is

$$E = h\nu = hc/\lambda = (6.63 \times 10^{-34}\ \text{J·s})(3 \times 10^{8}\ \text{m/s})/2.9 \times 10^{-7}\ \text{m}) = 6.86 \times 10^{-19}\ \text{J}$$
$$= (6.86 \times 10^{-19}\ \text{J})/(1.60 \times 10^{-19}\ \text{J/eV}) = 4.28\ \text{eV}$$

The maximum kinetic energy of the photoelectron is $E - W = 0.23$ eV, so the potential difference is 0.23 volts.

1-3 THE COMPTON EFFECT

Einstein's proposal that the quanta of energy should also manifest themselves outside of the "blackbody cavity," and the attendant explanation of the photoelectric effect, did not yet lead him to the notion of the quantum of radiation as a *particle*. It was not until 1916 that Einstein further analyzed the concept of a *gas of quanta* and was led to the prediction that the quanta, hereafter called **photons**, had to carry momentum as well as energy. The experiments that provided the most direct confirmation of the particle nature of radiation were carried out by Arthur H. Compton in 1923–4. Compton discovered that radiation of a given wavelength (in the X-ray region) sent through a metallic foil was scattered in a manner not consistent with classical radiation theory. According to classical theory, the mechanism for the effect is the re-radiation of light by electrons set into forced oscillations by the incident radiation, and this leads to the prediction of intensity observed at an angle θ that varies as $(1 + \cos^2\theta)$, and does not depend on the wavelength of the incident radiation. Compton found that the radiation scattered through a given angle actually consists of two components: one whose wavelength is the same as that of the incident radiation, the other of wavelength shifted relative to the incident wavelength by an amount that depends on the angle (Fig. 1-4). Compton was able to explain the "modified" component by treating the incoming radiation as a beam of photons of energy $h\nu$, with individual photons scattering elastically off individual electrons. In an elastic collision, momentum

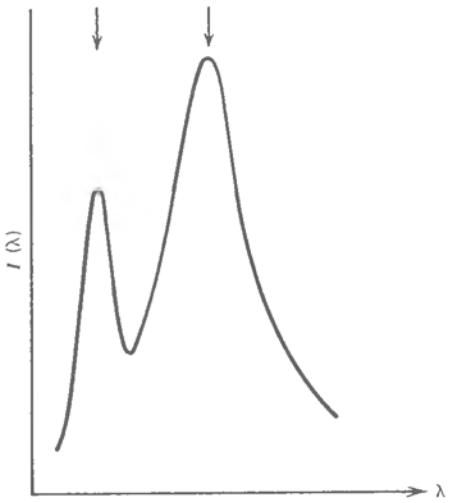


Figure 1-4 The spectrum of radiation scattered by carbon, showing the unmodified line at .07078 *nm* on the left and the shifted line at .07314 *nm* on the right. The former is the wavelength of the primary radiation.

as well as energy must be conserved, and we must first assign a momentum to the photon. By analogy with relativistic particle kinematics we argue that

$$p = \frac{h\nu}{c} \tag{1-10}$$

The argument is that it follows from the relativistic relation between energy and momentum

$$E = [(m_0c^2)^2 + (pc)^2]^{1/2} \tag{1-11}$$

where m_0 is the rest mass of the particle, that the velocity at this momentum is

$$v = \frac{dE}{dp} = \frac{pc^2}{E} = \frac{pc^2}{(m_0^2c^4 + p^2c^2)^{1/2}} \tag{1-12}$$

For a photon this is always c, and hence the *photon rest mass must be zero*. Thus the relation (1-11) becomes

$$E = pc \tag{1-13}$$

which yields (1-10) when we substitute $E = h\nu$. One may also derive (1-13) from consideration of the energy and momentum of an electromagnetic wave, but the analogy argument is simpler.

Consider, now, a photon with initial momentum $\mathbf{p}$ incident upon an electron at rest. After the collision, the photon momentum is $\mathbf{p}'$, and the electron recoils with momentum $\mathbf{P}$. Conservation of momentum yields (Fig. 1-5)

$$\mathbf{p} = \mathbf{p}' + \mathbf{P} \tag{1-14}$$

from which it follows that

$$\mathbf{P}^2 = (\mathbf{p} - \mathbf{p}')^2 = \mathbf{p}^2 + \mathbf{p}'^2 - 2\mathbf{p} \cdot \mathbf{p}' \tag{1-15}$$

Energy conservation reads

$$h\nu + m_ec^2 = h\nu' + (m_e^2c^4 + P^2c^2)^{1/2} \tag{1-16}$$

where m_e is the electron mass. When this is rearranged and squared, one finds that

$$\begin{aligned} m_e^2c^4 + P^2c^2 &= (h\nu - h\nu' + m_ec^2)^2 \\ &= (h\nu - h\nu')^2 + 2m_ec^2(h\nu - h\nu') + m_e^2c^4 \end{aligned}$$

On the other hand, (1-15) may be rewritten in the form

$$P^2 = \left(\frac{h\nu}{c}\right)^2 + \left(\frac{h\nu'}{c}\right)^2 - 2\frac{h\nu}{c} \cdot \frac{h\nu'}{c} \cos\,\theta$$

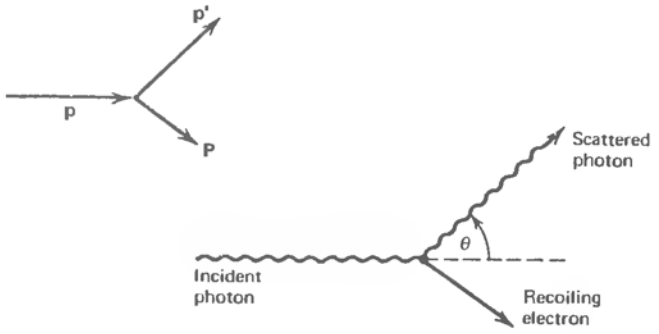


Figure 1-5 Kinematics for the Compton effect.

that is,

$$P^2c^2 = (h\nu - h\nu')^2 + 2(h\nu)(h\nu')(1 - \cos\theta) \tag{1-17}$$

where θ is the photon scattering angle. If we now insert P^2c^2 obtained from the energy conservation equation, we are led to

$$h\nu\nu'(1 - \cos\theta) = m_e c^2(\nu - \nu')$$

or equivalently

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta) \tag{1-18}$$

The measurements of the modified component agree very well with this prediction. The unmodified line is due to the scattering by the whole atom; if m_e is replaced by the mass of the atom, the shift in the wavelength is very small, since an atom is many thousand times more massive than an electron. The quantity $h/m_e c$ has the dimensions of a length. It is called the *Compton wavelength* of the electron, and its magnitude is

$$\frac{h}{m_e c} = 2.426 \times 10^{-12}\ \text{m} \tag{1-19}$$

Measurements of the electron recoil were also made, and these are in good agreement with the kinematic calculation outlined above. It was furthermore determined by good time resolution coincidence experiments that the outgoing photon and the recoil electron appear simultaneously. Thus there is no question of the correctness of the interpretation of the Compton effect as an elastic collision between a massive and a massless object. The particle-like behavior of the photon is not obvious to the naked eye. Nevertheless, it is interesting to note that although the energy of a quantum of radiation in the optical range, with $\lambda = 450$ nm, has a very tiny energy,

$$h\nu = hc/\lambda = (6.63 \times 10^{-34}\ \text{J·s})(3 \times 10^8\ \text{m/s})/(4.5 \times 10^{-7}\ \text{m}) = 4.24 \times 10^{-19}\ \text{J},$$

the dark-adapted eye can detect as few as 10 photons. A light bulb emitting 20 J/s of energy in that wavelength vicinity will emit

$$n = (20\ \text{J/s})/(4.24 \times 10^{-19}\ \text{J}) = 4.7 \times 10^{19}\ \text{photons/second}$$

The real difficulty arises from the knowledge that radiation also has wave properties and exhibits interference and diffraction. This gives rise to enormous conceptual difficulties, which will be addressed in Chapter 2; the resolution of these difficulties finally came with quantum mechanics.

EXAMPLE 1-5

A photon of energy 500 MeV strikes a proton at rest. What is its maximum energy loss?

SOLUTION The maximum momentum transfer to the photon occurs when the proton moves in the direction of the incident photon—a head-on collision. We can thus deal with this problem as a one-dimensional one. Let the initial photon momentum be $+q$, and let its final momentum be q'. Since the proton was initially at rest, its final momentum must be $q - q'$. Energy conservation gives us

$$qc + Mc^2 = |q'|c + \sqrt{(Mc^2)^2 + (q - q')^2c^2}$$